

Q1

A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop.

10 marks

A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop.

Write a Java program to:

Create an ArrayList of student names.

Read 8 student names from the user.

Traverse the list using an Iterator.

Display each student name on a separate line.

Display the total number of students stored in the list.

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 public class Main
4 {
5     public static void main(String[] args)
6     {
7         ArrayList<String> students = new ArrayList<String>();
8         students.add("Navin");
9         students.add("Dhanu");
10        students.add("Sona");
11        students.add("kamesh");
12        students.add("Hemala");
13        students.add("Dhandapani");
14        students.add("Jegadeshwari");
15        students.add("Sandhya");
16        Iterator<String> it = students.iterator();
17        System.out.println("Student Names:");
18        while(it.hasNext())
19        {
20            System.out.println(it.next());
21        }
22        System.out.println("Total Students="+students.size());
23    }
24 }
```

Input

stdin input

Output

```
Student Names:
Navin
Dhanu
Sona
kamesh
Hemala
```

Q2 Remove Even Numbers Using Iterator

10 marks

A company maintains a list of employee ID numbers. The management wants to remove all employees whose ID numbers are even, using an Iterator.

Write a Java program to:

Create an ArrayList<Integer>.

Read 10 employee ID numbers from the user.

Traverse the list using an Iterator.

Remove all even ID numbers using the remove() method of the Iterator.

Display the updated list containing only odd employee IDs.

Main.java



Share

Run

```
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args)
5     {
6         ArrayList<Integer> emp = new ArrayList<Integer>();
7         emp.add(101);
8         emp.add(102);
9         emp.add(103);
10        emp.add(104);
11        emp.add(105);
12        emp.add(106);
13        emp.add(107);
14        emp.add(108);
15        emp.add(109);
16        emp.add(110);
17        Iterator<Integer> it = emp.iterator();
18        while (it.hasNext())
19        {
20            int id = it.next();
21            if (id % 2 == 0)
22            {
23                it.remove();
24            }
25        }
```

```
        System.out.println("Employee IDs after removing even
        numbers:");
        Iterator<Integer> it2 = emp.iterator();
        while (it2.hasNext())
        {
            System.out.println(it2.next());
        }
    }
}
```

Output

Employee IDs after removing even numbers:

101

103

105

107

109

Q3 Product Price Analysis Using Iterator

10 marks

A supermarket stores the prices of various products in an `ArrayList<Double>`. The manager wants to analyze the prices using an Iterator.

Write a Java program to:

Create an `ArrayList<Double>`.

Read the prices of 10 products from the user.

Traverse the list using an Iterator.

Calculate and display:

Total price of all products.

Average product price.

Highest product price.

Display all product prices using the Iterator.

Your Code

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         ArrayList<Double> prices = new ArrayList<Double>();
7         prices.add(100.0);
8         prices.add(250.0);
9         prices.add(300.0);
10        prices.add(150.0);
11        prices.add(400.0);
12        prices.add(500.0);
13        prices.add(350.0);
14        prices.add(450.0);
15        prices.add(200.0);
16        prices.add(600.0);
17        double total = 0;
18        double highest = 0;
19        Iterator<Double> it = prices.iterator();
20        System.out.println("Product Prices:");
21        while(it.hasNext())
22        {
23            double price = it.next();
24            System.out.println(price);
25            total = total + price;
26            if(price > highest)
```

```
        if(price>highest)
        {
            highest=price;
        }
    }
    double average=total/prices.size();
    System.out.println("\n Total Price="+total);
    System.out.println("Average Price="+average);
    System.out.println("Highest Price="+highest);
}
}
```

Product Prices:

100.0
250.0
300.0
150.0
400.0